

REPORT

on

Post Lab

Submitted in partial fulfilment

of Subject

Image Processing Lab (N-PECCS502P)

(V Semester, B.Tech. CSE)

(ACADEMIC SESSION 2026-27)

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NAGPUR UNIVERSITY, NAAC ACCREDITED WITH 'A' GRADE)**

POST LAB DETAILS

USN No.	CS24099
STUDENT NAME	Pranay Aknurwar
POST LAB	Create a website for Image Processing techniques. (Include all 12 practical's)
GitHub REPOSITORY URL:	
GitHub REPOSITORY URL QR CODE:	
DEPLOYMENT URL:	
DEPLOYMENT URL QR CODE:	

1. ABSTRACT

The Image Processing Lab Web Application is a full-stack web-based system developed to provide an interactive platform for performing and demonstrating various fundamental image processing techniques. The project is designed for an Image Processing Laboratory, where students can perform practical experiments through a user-friendly web interface instead of relying on separate programs for each experiment.

The system implements a collection of image processing operations including RGB and grayscale conversion, arithmetic and bitwise operations, geometric transformations, histogram equalization, smoothing, sharpening, thresholding, spatial filtering, image inpainting, lossless compression, morphological operations, correlation-based object detection, colour-space conversion, and edge detection using Canny, Sobel, and Prewitt methods. Users can select or upload an image and apply the required operation through separate practical tabs.

The application follows a full-stack architecture consisting of a frontend, backend, and database. The frontend is developed using HTML, CSS, JavaScript, and OpenCV.js, providing the interactive interface and performing image-processing operations in the browser. The backend is developed using Java and Spring Boot, which provides REST APIs for student information, experiment information, and experiment execution history. PostgreSQL is used as the relational database, with Spring Data JPA/Hibernate used for database communication and storage of experiment records.

The main objective of the project is to combine theoretical image processing concepts with practical implementation in a centralized and accessible environment. The system improves the learning experience by allowing students to visualize the results of different algorithms immediately, while the backend and database provide a structured mechanism for maintaining experiment information and activity records. The project demonstrates the integration of image processing, web development, REST APIs, Java, OpenCV, and relational database management in a single educational application.

2. INTRODUCTION

Image processing is the process of improving, changing, and analyzing digital images using computer techniques. It is widely used in areas such as medical imaging, security, photography, and computer vision.

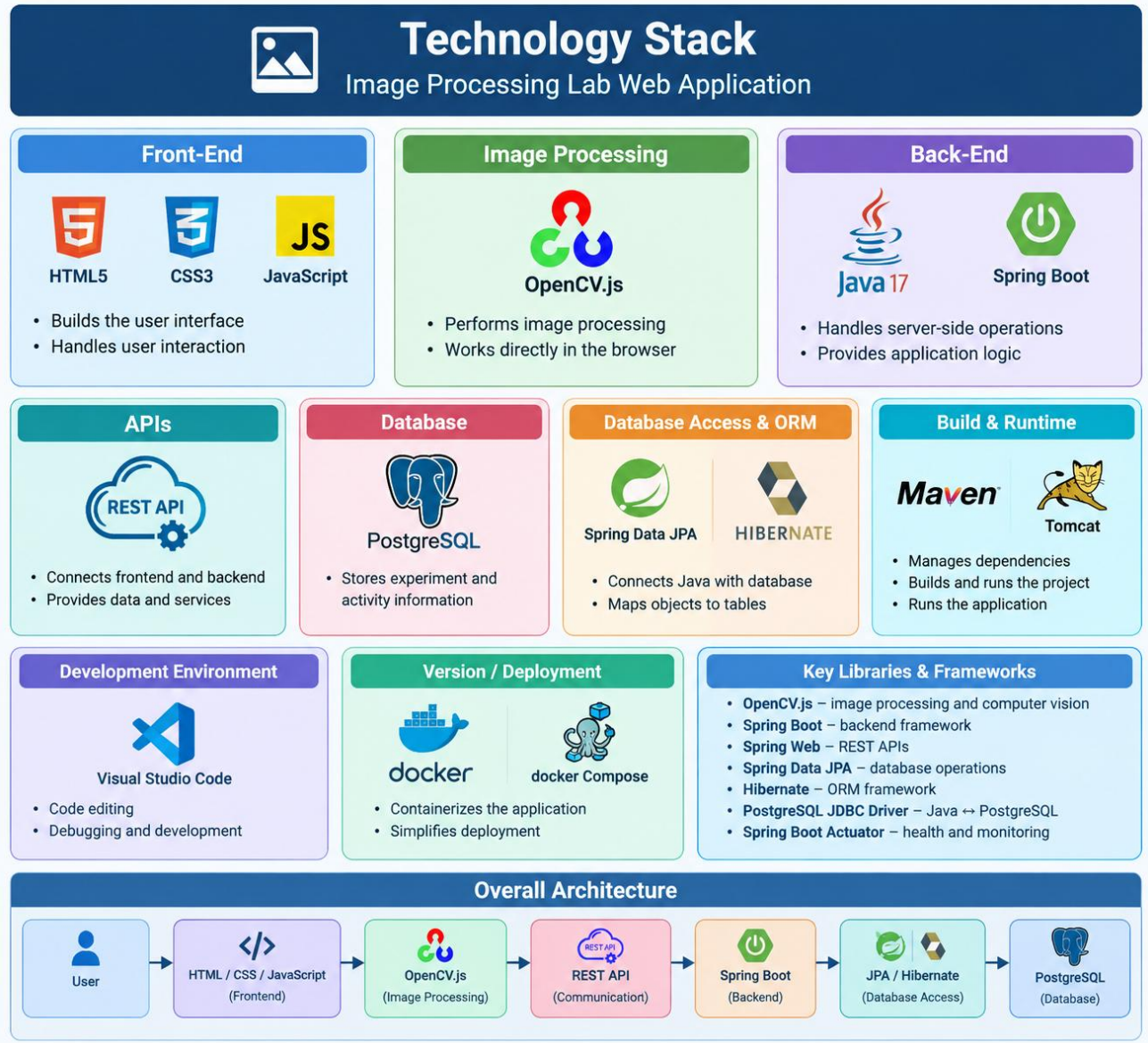
The Image Processing Lab Web Application is developed to provide students with a simple platform for performing different image processing practicals. Users can upload an image, select an experiment, apply different operations, and view the result directly on the website.

The project includes practicals such as image conversion, geometric transformations, image enhancement, filtering, inpainting, compression, morphological operations, object detection, colour-space conversion, and edge detection.

The application uses HTML, CSS, and JavaScript with OpenCV.js for the frontend. A Java Spring Boot backend provides APIs for the application, while PostgreSQL is used to store experiment-related information and activity records.

The main purpose of this project is to make image processing experiments simple, interactive, and easy to understand through a single web-based platform.

3. TECHNOLOGY STACK



4. APPLICATION SCREENSHOTS

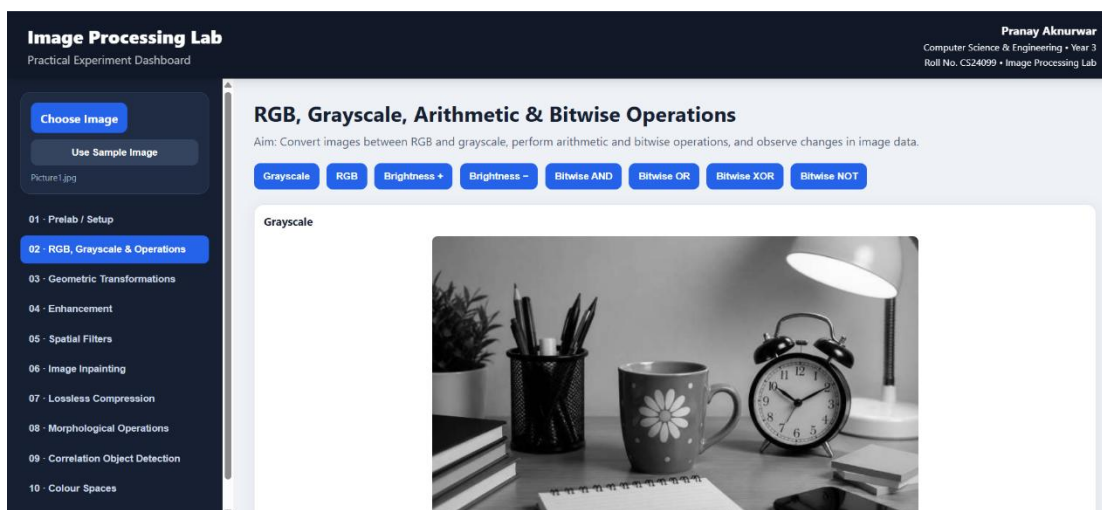


Figure 1 : Practical 1

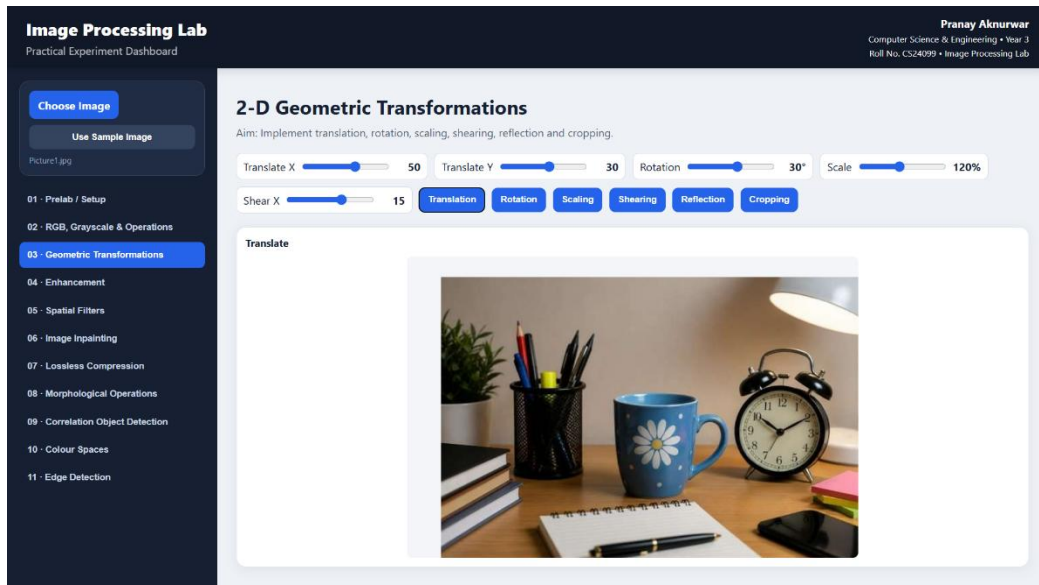


Figure 2 : Practical 2



Figure 3 : Practical 3

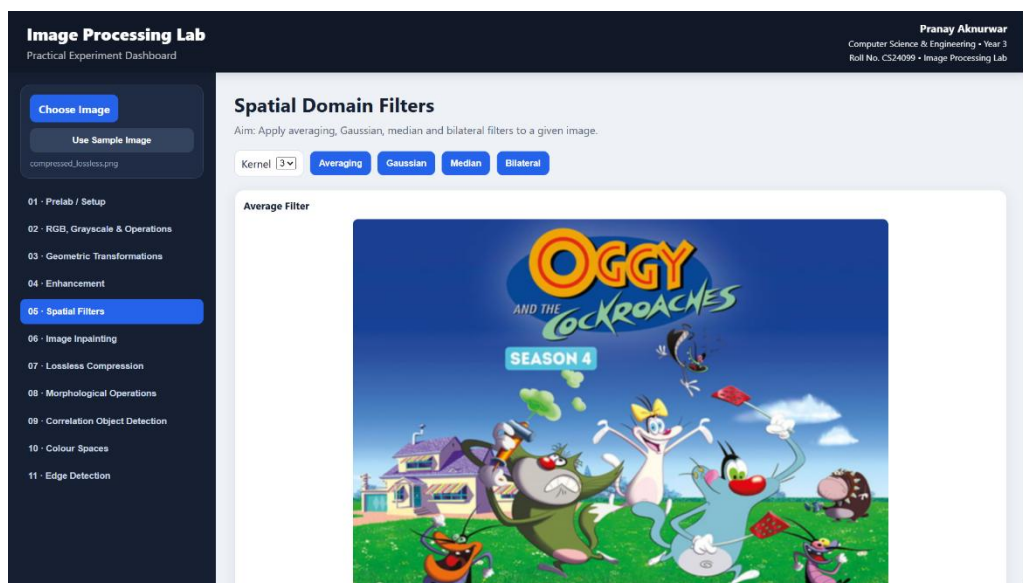


Figure 4 : Practical 4

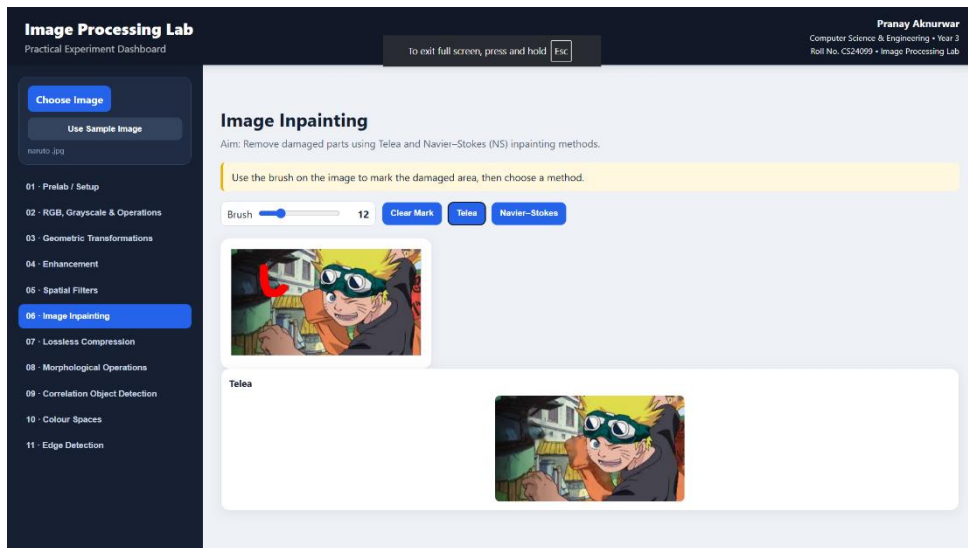


Figure 5 : Practical 5

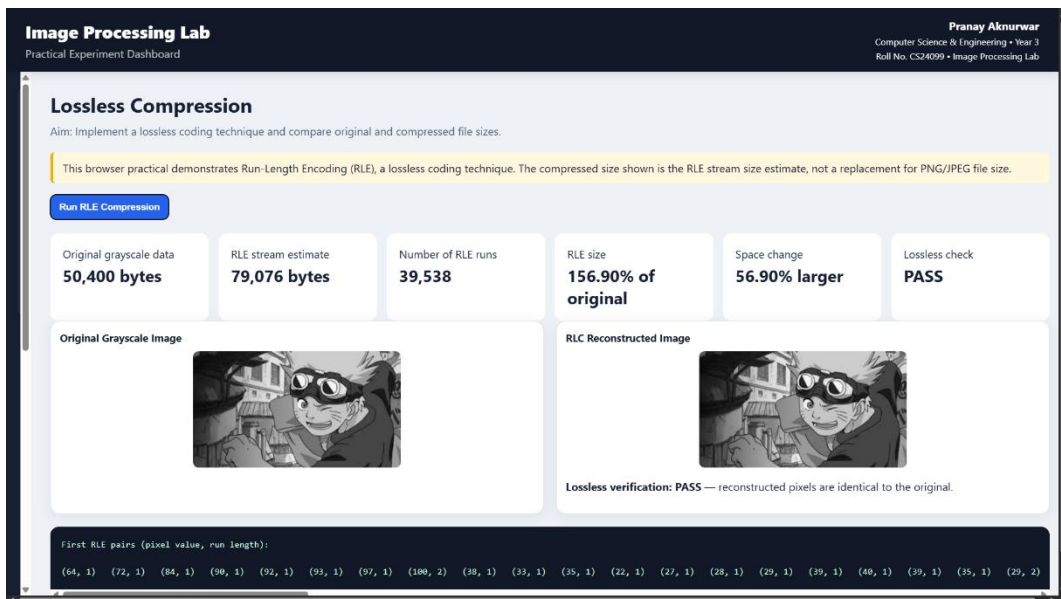


Figure 6 : Practical 6

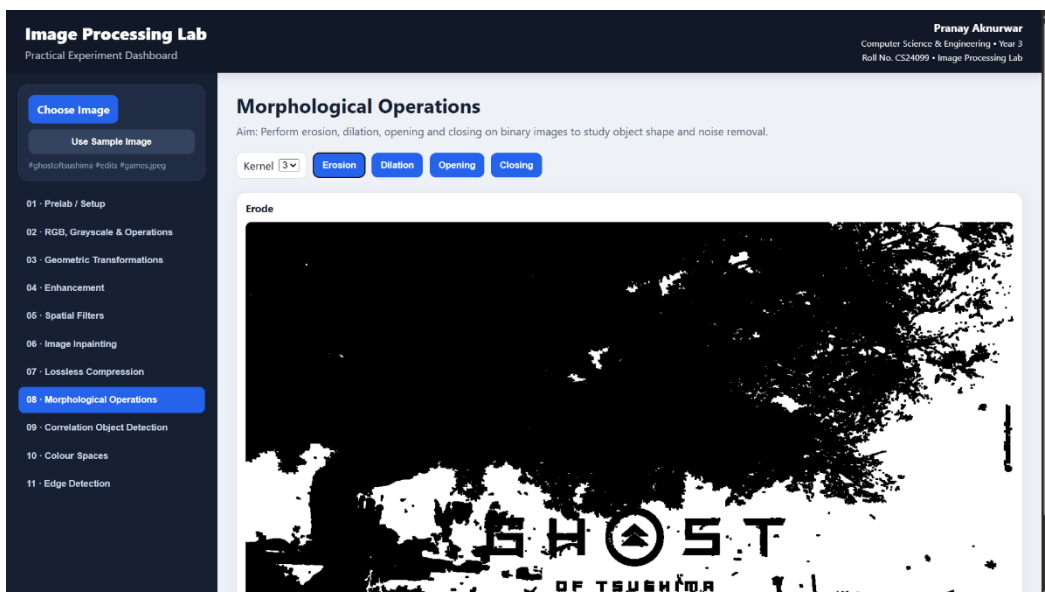


Figure 7 : Practical 7

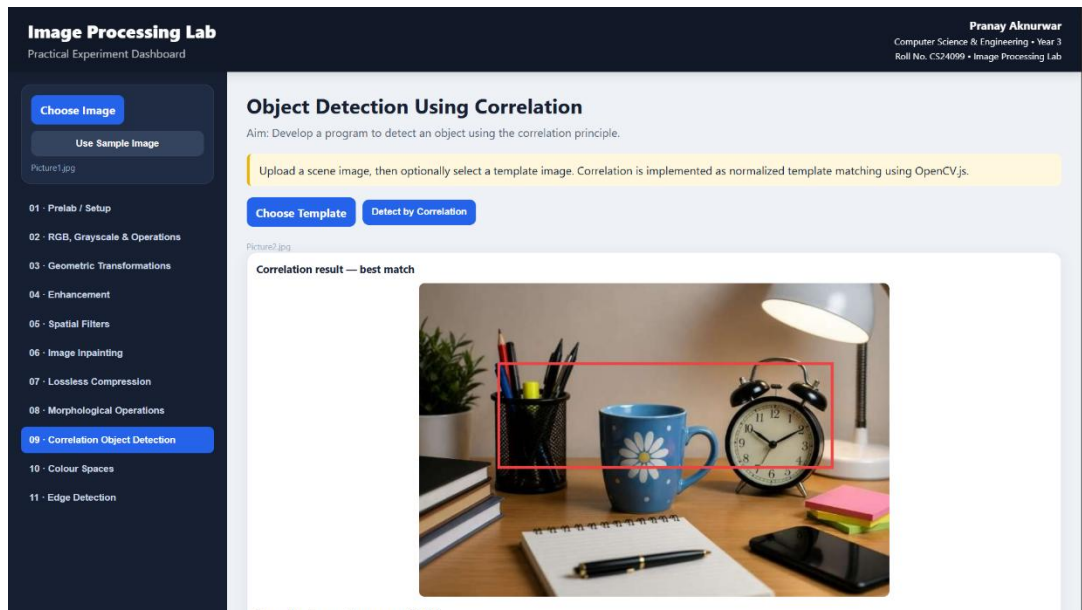


Figure 8 : Practical 8

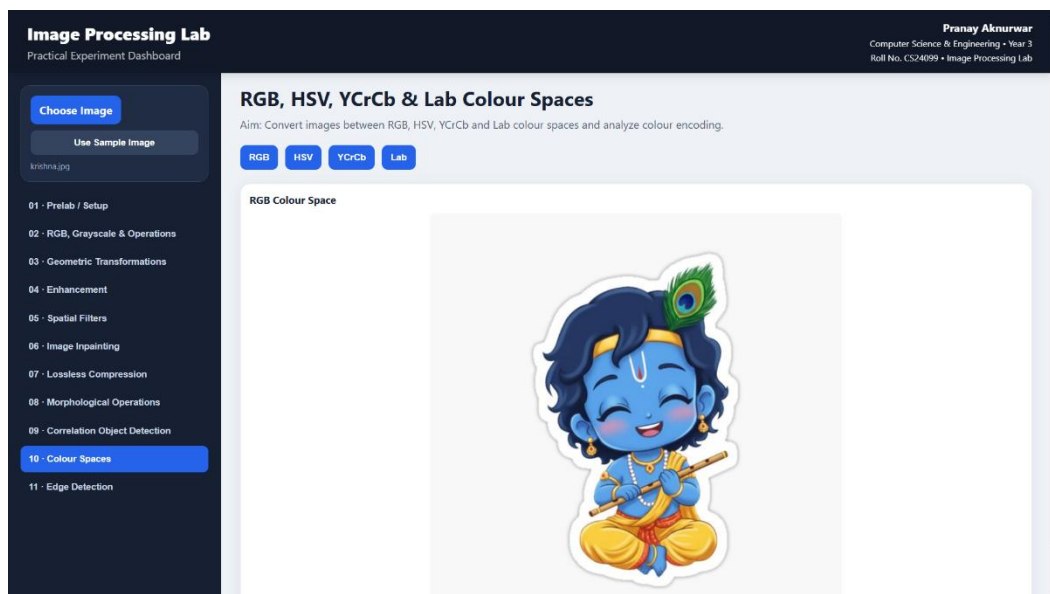


Figure 9 : Practical 9

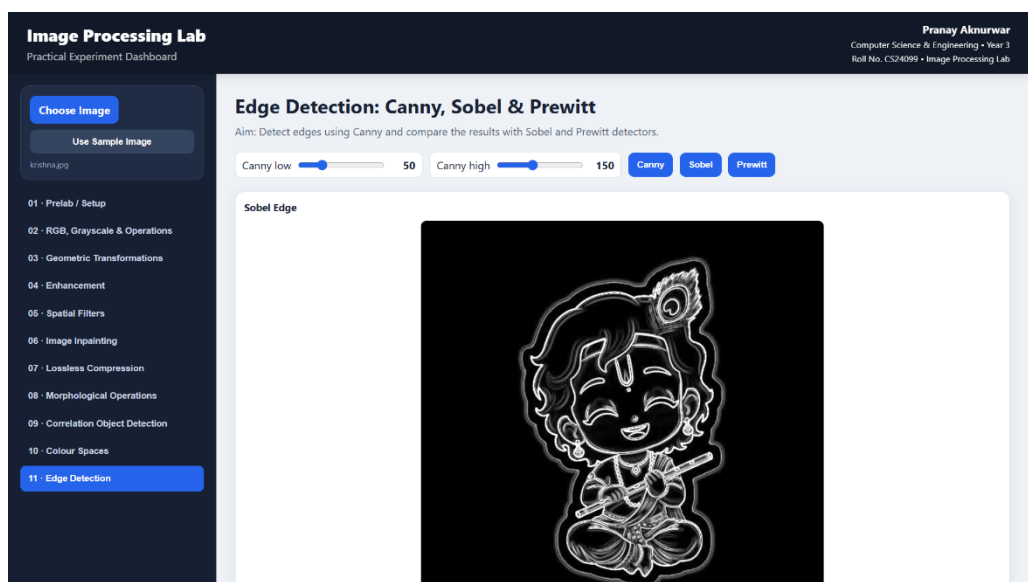


Figure 10 : Practical 10

5. CONCLUSION

The **Image Processing Lab Web Application** successfully provides an interactive platform for performing and understanding various image-processing operations through a web browser. The project integrates **HTML, CSS, JavaScript, and OpenCV.js** on the frontend with **Java 17, Spring Boot, REST APIs, and PostgreSQL** on the backend.

The application demonstrates practical image-processing techniques such as image enhancement, filtering, segmentation, inpainting, run-length coding, morphology, correlation, color-space conversion, and edge detection. The use of OpenCV.js enables image processing directly in the browser, while the Spring Boot backend provides a structured and scalable architecture for storing experiment-related information.

Overall, the project combines **image processing concepts with full-stack web development**, making laboratory experiments more interactive, accessible, and easier to understand. It also provides a strong foundation for extending the system with additional image-processing algorithms, user authentication, experiment history, performance analysis, and advanced computer-vision features.

6. REFERENCES

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